

Matrix functions and stability

Degree coverage with seven matrix multiplications

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved

Rating rationale: Historical ratings for the original conjecture: challenging because numerical evidence for coefficient-space coverage needs exact algebraic justification; community impact comes from multiplication budgets for practical matrix-function evaluation.

Resolution — exact maximum 47, 17 September 2026

Author: Marcus Webb, The University of Manchester. Developed with substantial ChatGPT/Codex assistance. Separate Codex agents reviewed the mathematics, and an independently implemented exact verifier reconstructed the full polynomial and its derivatives. This is informal AI-agent review, not external human peer review or formal verification.

[Theorem 1 of the complete proof](#) determines the exact maximum in the retained question:

$$d_7 = 47.$$

Every complex polynomial of degree at most 47 belongs to the Zariski closure of seven-product outputs in the full coefficient space through degree 128. An explicit seven-product family has a rigorously certified order-47 contact with an invertible 48-by-48 coefficient Jacobian. Exact Gaussian-rational contraction bounds prove the contact exists; the inverse function theorem and a weighted input/output limit give the universal coverage statement, with every higher coefficient tending to zero.

A self-contained available-space dimension count gives $\dim \overline{\mathcal{P}_7}^Z \leq 49$. Irreducibility and the seven-product polynomial x^{128} then exclude coverage of the entire degree-at-most-48 space. The new proof does not depend on the earlier border construction or on an external dimension theorem. Exact representation of every degree-47 polynomial and numerically stable coefficient recovery are not asserted.

[Standalone proof source](#) · [Exact certificates and reproduction](#) · [Independent review](#). The original equality $d_7 = 42$, permanent ID and canonical path remain unchanged. The earlier degree-42 and degree-44 results below retain their attribution and archived evidence.

Earlier negative resolution — degree 44, 17 September 2026

Author: Marcus Webb, The University of Manchester. Developed with substantial ChatGPT assistance; two separate Codex agents independently reviewed the proof and recomputed its exact certificates. This is informal AI-agent review, not external human peer review or formal verification.

The conjectured equality is false. [Theorem 1 of the complete proof](#) establishes

$$\mathbb{C}[x]_{\leq 44} \subseteq \overline{\mathcal{P}_7}^Z.$$

The construction first retains a suitable quadruple of polynomials in the joint closure of four-product computations (Lemma 2). Three more products give a 45-parameter coefficient map whose exact Jacobian determinant is 256. The argument uses the original complex field and all coefficients through degree 128; high-degree coefficients vanish in the limit rather than being discarded.

Corollary 3, using Jarlebring–Lorentzon’s dimension theorem, gives $44 \leq d_7 \leq 47$ for the maximum in the retained question. That contribution left the exact maximum undetermined while giving a complete negative answer to the specific equality $d_7 = 42$; the new theorem above now determines it as 47. No

exact representation of every degree-44 polynomial, real Euclidean density, or numerical stability is asserted.

[Proof source](#) · [Independent reviews and reproduction](#) · [Source conversation and attribution](#). The permanent ID, canonical path and original mathematical target are retained. The degree-42 result below retains its separate credit.

Verified partial result — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the stated partial results.

A fixed seven-product scheme has a full-rank complex coefficient map, certified by a nonzero exact integer Jacobian minor. Its image contains a nonempty Zariski-open subset of $\mathbb{C}[x]_{\leq 42}$ and is Euclidean dense there.

This earlier result did not prove an upper bound excluding degree 43 and above, so at the time it left the maximal-degree equality open. It asserted neither exact representation of every polynomial nor real Euclidean dense coverage. The later degree-44 construction above now refutes that equality; the degree-42 theorem remains valid.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1 and equations (1)-(2)**. [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

Start with the scalar polynomials $1, x \in \mathbb{C}[x]$. Allow arbitrarily many complex linear combinations of already computed polynomials and at most seven multiplications of two already computed polynomials. Let $\mathcal{P}_7 \subseteq \mathbb{C}[x]_{\leq 128}$ be the set of possible outputs.

Identify a polynomial with its coefficient vector in $\mathbb{C}^{129}$ and let $\overline{\mathcal{P}_7}^Z$ be its Zariski closure: the common zero set of all polynomial equations in the coefficients that vanish throughout $\mathcal{P}_7$. Is

$$\max\{d \in \{0, \dots, 128\} : \mathbb{C}[x]_{\leq d} \subseteq \overline{\mathcal{P}_7}^Z\} = 42?$$

The target concerns closure and the complex coefficient field. Exact representability of every polynomial is a stronger assertion.

Why it matters

For a dense input matrix, matrix products dominate the cost of polynomial evaluation. The conjecture determines how much polynomial degree a budget of seven products can cover, providing a theoretical benchmark for matrix-function evaluation schemes.

References

1. E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3 (13 August 2025), §2.2 (closure convention), §5.4, Conjecture 13. [Primary text](#).
2. J. Sastre, J. Ibáñez, J. M. Alonso, and E. Defez, *Beyond Paterson–Stockmeyer: Advancing matrix polynomial computation*, WSEAS Transactions on Mathematics 24 (2025), 684–693, §4 (a five-product construction). [Primary paper](#), [DOI](#).
3. J. M. Alonso, J. Sastre, J. Ibáñez, and E. Defez, *A systematic framework for stable and cost-efficient matrix polynomial evaluation*, arXiv:2603.23143v1 (24 March 2026), abstract and conclusions. [Primary text](#).

Status check — 2026-09-08

The latest arXiv source remains v3 and labels the displayed equality a conjecture supported by computations. The later papers give constructions and stability procedures without establishing the seven-product degree-coverage claim. Searches included the exact title with 2026, *Jarlebring Lorentzon conjecture seven multiplications 42*, and *Sastre polynomial evaluation 2026*. No resolution was located. A [3 September 2026 author talk](#) also describes higher-cost degree coverage as conjectural. Six-product real/complex formulations have a separate source ambiguity and are deliberately not included here.

Audit — 2026-09-10

Rechecked [Conjecture 13](#) and [Remark 14](#): complex closure coverage at seven products is still conjectural, with concrete matrix-exponential applications. Title and seven-product searches found no exact proof. The [June 2026 degree-eight construction](#) solves a different budget/degree problem.